

A minor correction to Anne Mangel's "Maxwell's Demon, Entropy, Information: The Crying of Lot 49"

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This is a brief, and perhaps pedantic, post to bring attention to an essay of extreme prominence in Pynchon studies, Anne Mangel, "Maxwell's Demon, Entropy, Information: The Crying of Lot 49", *Triquarterly* 20 (1971): 194-208. (Future reference will be to the reprinted version Mangel, Anne. "Maxwell's Demon, Entropy, Information: The Crying of Lot 49." In *Mindful Pleasures: Essays on Thomas Pynchon*, edited by George Levine and David Leverenz, 87-99. Boston: Little Brown, 1976, pp. 87-99.)

At the beginning of the essay, Mangel states:

James Clerk Maxwell introduced the Demon in 1871 in his book Theory of Heat. (Ibid., p. 87)

This is not, as my recent research has revealed, strictly correct. Instead, as Cargill Gilston Knott relates, the Demon was first formalized in a letter from Maxwell to Tait on December 11th, 1867, the contents of which was as follows:

"I do not know in a controversial manner the history of thermodynamics, that is, I could make no assertions about the priority of authors without referring to their actual works... Any contributions I could make to that study are in the way of altering the point of view here and there for clearness or variety, and picking holes here and there to ensure strength and stability.

"As for instance I think that you might make something of the theory of the absolute scale of temperature by reasoning pretty loud about it and paying it due honour at its entrance. To pick a hole -- say in the 2nd law of thermodynamics, that if two things are in contact the hotter cannot take heat from the colder without external agency.

"Now let A and B be two vessels divided by a diaphragm and let them contain elastic molecules in a state of agitation which strike each other and the sides.

"Let the number of particles be equal in A and B but let those in A have the greatest energy of motion. Then even if all the molecules in A have equal velocities, if oblique collisions occur between them their velocities will become unequal, and I have shown that there will be velocities of all magnitudes in A and the same in B, only the sum of the squares of the velocities is greater in A than in B.

"When a molecule is reflected from the fixed diaphragm CD no work is lost or gained.

"If the molecule instead of being reflected were allowed to go through a hole in CD no work would be lost or gained, only its energy would be transferred from the one vessel to the other.

"Now conceive a finite being who knows the paths and velocities of all the molecules by simple inspection but who can do no work except open and close a hole in the diaphragm by means of a slide without mass.

"Let him first observe the molecules in A and when he sees one coming the square of whose velocity is less than the mean sq. vel. of the molecules in B let him open the hole and let it go into B. Next let him watch for a molecule of B, the square of whose velocity is greater than the mean sq. vel. in A, and when it comes to the hole let him draw the slide and let it go into A, keeping the slide shut for other molecules.

"Then the number of molecules in A and B are the same as at first, but the energy in A is increased and that in B is diminished, that is, the hot system has got hotter and the cold colder and yet no work has been done, only the intelligence of a very observant and neat-fingered being has been employed.

"Or in short if heat is the motion of finite portions of matter and if we can apply tools to such

portions of matter so as to deal with them separately, then we can take advantage of the different motion of different proportions to restore a uniformly hot system to unequal temperatures or to motions of large masses.

*"Only we can't, not being clever enough." (Maxwell, James Clerk. "Letter from Maxwell to Tait, Dec. 11, 1867." In *Life and Scientific Work of Peter Guthrie Tait*, edited by Cargill Gilston Knott, 3:213-214. Cambridge: Cambridge University Press, 1911.)*

Finally, it is worth quoting another document here that should be of interest to Pynchon scholars, that is the humorous catechism written by Maxwell to sum up the Demon:

"Concerning Demons.

"1. Who have them this name? Thomson.

"2. What were they by nature? Very small BUT lively beings incapable of doing work but able to open and shut valves which move without friction or inertia.

"3. What was their chief end? To show that the 2nd Law of Thermodynamics has only a statistical certainty.

*"4. Is the production of the inequality of temperature their only occupation? No, for less intelligent demons can produce a difference in pressure as well as temperature by merely allowing all particles going in one direction while stopping all those going the other way. This reduces the demon to a valve. As such value him. Call him no more a demon but a valve like that of the hydraulic ram, suppose. (Maxwell, James Clerk. "Undated letter from Maxwell to Tait." In *Life and Scientific Work of Peter Guthrie Tait*, edited by Cargill Gilston Knott, 3:214-215. Cambridge: Cambridge University Press, 1911.)*

It's possible that somebody else in Pynchon work has picked up on these sources, but I thought I'd set that straight and cite here.

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